1.Variables in Java

A variable is a named memory location that stores a value which can change while the program runs. In Java, every variable must have a data type declared before use.

Types of Variables

Java has three kinds of variables, based on where they are declared.

1. Local Variables

Declared inside a method, constructor, or block. They exist only while that method runs and must be initialized before use.

2. Instance Variables

Declared inside a class but outside any method, without static. Each object gets its own copy. They get default values (0, null, false) if not initialized.

3. Static (Class) Variables

Declared with the static keyword. Only one copy is shared across all objects of the class.

Program For Variables.

```
public class Student {

    // Instance variables
   String name;
   int age;
    // Static variable
   static String college = "ABC College";
 void display() {
     // Local variable
     String message = "Student Details";

     System.out.println(message);
```

```
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("College: " + college);
    }

    public static void main(String[] args) {
        Student s1 = new Student();
        s1.name = "Akash";
        s1.age = 22;

        Student s2 = new Student();
        s2.name = "Rahul";
        s2.age = 21;

        s1.display();
        System.out.println("---");
        s2.display();

        // Changing static variable affects all objects
        Student.college = "K.J Somaiya College Kopargaon";
        s1.display();
    }
}
```

2. Operators In Java

Operators perform operations on variables and values.

Category	Operators	Example
Arithmetic	+ - * / %	a + b, 10 % 3 (gives 1)
Relational	== != > < >= <=	a > b
Logical	&& || !	(a > 5 && b < 10)
Assignment	= += -= *= /= %=	a += 5 (same as a = a + 5)
Unary	++ -- + - !	a++, --b
Ternary	? :	(a > b) ? a : b
Bitwise	& | ^ ~ << >>	a & b

Program For Operators

```
public class OperatorsDemo {
public static void main(String[] args) {
        int a = 10, b = 3;
        // Arithmetic
        System.out.println("Sum: " + (a + b));
        System.out.println("Division: " + (a / b));
        System.out.println("Remainder: " + (a % b));
       // Relational
        System.out.println("a > b: " + (a > b));
      // Logical
        System.out.println(a > 5 && b < 2);
        System.out.println(a > 5 || b < 2);
     // Unary
        int x = 5;
```

```
        System.out.println(x++);
        System.out.println(++x);
       // Ternary
        int max = (a > b) ? a : b;
        System.out.println("Max: " + max);          // 10
    }
}
```

3. Conditional Statement

They let the program make decisions.

if, if-else, else-if ladder

Program For Conditional Statement.

if Condition

```
class Main {
  public static void main(String args[])
    int marks = 72;

    if (marks >= 90) {
       System.out.println("Grade A+");
    } else if (marks >= 75) {
       System.out.println("Grade A");
    } else if (marks >= 60) {
       System.out.println("Grade B");
    } else {
       System.out.println("Grade C");
    }
    }
```

Nested if Condition

```
Class Main{
Public static void main(String args[])
```

```
int age = 20;
boolean hasID = true;

if (age >= 18) {
   if (hasID) {
      System.out.println("Allowed to vote");
   }
}
}
```

switch

Best when comparing one variable against many fixed values.

Program For Switch Statement.

```
Class Main{
Public static void main(String args[])
int day = 3;

switch (day) {
   case 1:
      System.out.println("Monday");
      break;
   case 2:
      System.out.println("Tuesday");
      break;
   case 3:
      System.out.println("Wednesday");
      break;
   default:
      System.out.println("Invalid day");
}
}
```

4.Loop In java

. for loop

Use when the number of iterations is known.